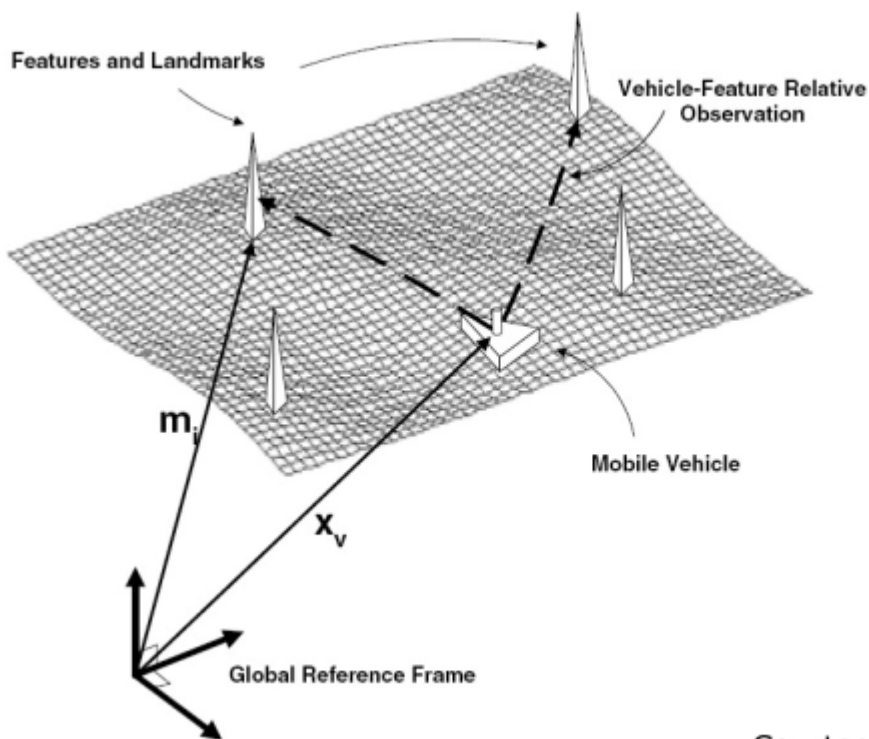
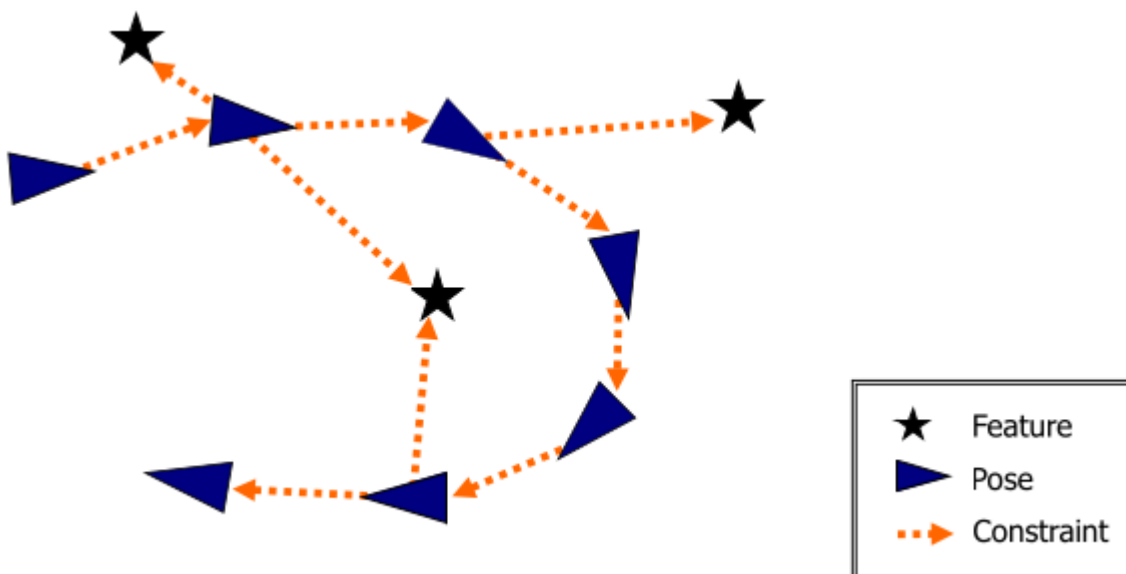


L18: Graph SLAM Landmarks



Courtesy: ACFR



- Nodes can represent:
 - Robot poses
 - Landmark locations
- Edges can represent:
 - Landmark observations
 - Odometry measurements
- Minimisation optimises landmark location and robot poses jointly

Landmark Observation

- Landmark representation

- (x,y) point in world
- Relative observation in plane (x,y)

- **Expected observation**(x-y sensor)

- $\hat{z}_{ij}(\underset{\text{robot}}{x_i}, \underset{\text{landmark}}{x_j}) = R_i^T(x_j - \underset{\text{robot translation}}{t_i})$
- Error Function

$$\begin{aligned} e_{ij}(x_i, x_j) &= \hat{z}_{ij} - z_{ij} \\ &= R_i^T(x_j - t_i) - z_{ij} \end{aligned}$$

- **Bearing Only Observation**

- Landmark is still 2D point
- Robot observes only bearing toward landmark
- Observation function:
- $\hat{z}_{ij}(\underset{\text{robot}}{x_i}, \underset{\text{landmark}}{x_j}) = \arctan \frac{(x_j - t_i) \cdot y}{(x_j - t_i) \cdot x} - \underset{\text{robot-landmark angle}}{\theta_i} \underset{\text{robot orientation}}{}$
- Error function:
- $e_{ij}(x_i, x_j) = \frac{(x_j - t_i) \cdot y}{(x_j - t_i) \cdot x} - \theta_i - z_j$

Rank of Matrix H

- Rank of H_{ij} for **2D landmark pose** constraint

- Blocks of J_{ij} are at most 2×5 matrices
- H_{ij} cannot have more than rank 2
 - $\text{rank}(A^T A) = \text{rank}(A^T) = \text{rank}(A)$

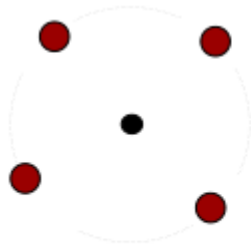
- Rank of H_{ij} for **Bearing-only** constraint

- Blocks of J_{ij} are at most 1×5 matrices
- H_{ij} has rank 1

Robot location

- Robot observes one landmark (x,y)

- Robot relative to landmark

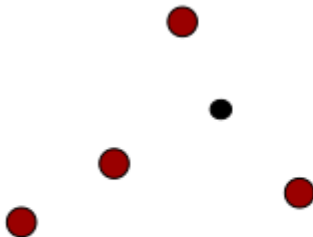


The robot can be somewhere on a circle around the landmark

It is a 1D solution space (constrained by the distance and the robot's orientation)

■

- Bearing only



The robot can be anywhere in the x-y plane

It is a 2D solution space (constrained by the robot's orientation)

○

Rank

- In landmark-based SLAM, system likely to be under-determined
- Rank $H \leq$ sum of ranks of constraints
- For **unique solution**: System must have **full rank**

Under-Determined Systems

- No guarantee for full rank system
 - Landmarks may be observed only once
 - Robot might have no odometry
- We can still deal with these situation by adding a "damping" factor to H
- Instead of solving $H\Delta x = -b$, we solve
 - $(H + \lambda I)\Delta x = -b$
 - **Damping Factor λI**
 - Make system positive definite
 - Weighted sum of Gauss Newton and steepest descent
 - Damping to regulate the convergence using backup/restore actions

Simplified Levenberg Marquardt

```
x: the initial guess
while (! converged)
     $\lambda = \lambda_{init}$ 
    <H,b> = buildLinearSystem(x) ;
    E = error(x)
    xold = x;
     $\Delta x$  = solveSparse( (H +  $\lambda$  I)  $\Delta x$  = -b) ;
    x +=  $\Delta x$ ;
    If (E < error(x)) {
        x = xold;
         $\lambda$  *= 2;
    } else {  $\lambda$  /= 2; }
```

Strong Relation: Bundle Adjustment

- 3D reconstruction based on images taken at different viewpoints
- Minimises the reprojection error in 2D image planes
- No notation of odometry (pose-pose)
- Often uses Levenberg Marquardt
- Developed in photogrammetry during the 1950ies